

Prediction of Crop Yield Based-on Soil Moisture using Machine Learning Algorithms

This work presents a smart farming intelligence platform that integrates low-cost IoT sensing, long-range LoRa communication, local edge processing, and machine-learning-driven crop and irrigation guidance tailored for rural, low-connectivity contexts. Field nodes (ESP32-based) collect soil and environmental parameters (soil moisture, N, P, K, pH, temperature/humidity) and transmit them via LoRa to a gateway which performs local logging (SD card), on-site analytics and automated farmer alerts via GSM/SMS. A machine-learning module—demonstrated experimentally with decision-tree models on soil nutrient and moisture features—produces crop recommendations and yield predictions; preliminary results in the study show high apparent accuracy on a limited dataset (paper-reported $\approx 95\%$), indicating feasibility but requiring larger-scale validation. The platform design (multilingual AI assistant, farmer mobile app, government analytics dashboard) emphasizes usability, scalability and policy-level integration for irrigation planning and subsidy verification. Key contributions are the hybrid architecture suited for connectivity-challenged regions, coupling edge intelligence with long-range, low-power communications, and a practical product ecosystem for farmer adoption. Remaining work includes robust model validation with larger, geographically diverse datasets, field pilots to quantify yield and water-use benefits, sensor calibration and hardware hardening, and a clarified sustainable business model for wide deployment.

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Abstract - Agriculture planning is playing an important role as the economic growth is very high day by day in our daily life. There is lot of research study going on as there are few important issues like soil nutrients, crop prediction, farming system, crop monitoring in agriculture with modern farming system. Crop prediction and crop monitoring is main factor to produce good quality of crops for farmers to predict crop yield based on soil moisture. Prediction of crop yield includes forecasting factors like temperature, humidity, rainfall, etc., and crop yield based on soil moisture includes few measures like NPK (Nitrogen, Phosphorous and potassium) and pH values using various sensors. Machine learning (ML) is a useful decision-making model for estimating crop yields, and also for deciding what crops to plant and what to do during the crop's growing seasons. To aid agricultural yield prediction studies, a number of analytical techniques have been used. In this study Farmers can predict or come to a decision the type of soil moisture values; Farmers can choose the type of crop they want to sow. In this paper, Author proposed decision tree supervised machine learning algorithm to improve the results for the prediction of crop yield based on soil moisture parameters to achieve better error rate and accuracy for economic growth. It also includes the few machine learning algorithms which are discussed in literature survey, further Author highlighted the proposed system in methodology, and compared the analysis in results to give it a balance view. The future scope is also mentioned to improve it for further studies. This paper will be sufficient for those who are keener in learning about the expectation of crop yield based on soil moisture using ML Algorithms.

Keywords - Agriculture, Crop Prediction, Machine learning, Soil Moisture, Yield

I. INTRODUCTION

Agriculture is the mainstay of commerce and the economic foundation of many nations. Farmers must be prepared with contemporary farming systems as traditional farming practices provide many constraints in today's world

of agriculture. For the improvement of production of crop, soil moisture data are used. Soil moisture nutrients are essential for better enhancement of crop growth and plant tissues [1]. Crop production prediction which maximizes the rate of crop production is one of the interesting research work done by increasing the soil moisture data to give efficient and accurate results. Crop prediction depends on important parameters of a region like – many harvesting techniques, weather conditions, soil moisture type (such as clay, sandy, saline soil, etc), soil composition (such as pH value, potassium, nitrogen, phosphorous, etc) and many more [2], [3]. Various ML methods may be used to improve or interpret the soil moisture information. The technology, which was developed using a ML algorithm, forecasts the best crops for farmers based on the region's soil condition. “Artificial neural networks” (ANN) for wheat yield prediction using linear regression for crop yield goal setting, meteorological data, support vector machines for crop yield prediction, and other machine learning techniques are available for use in agricultural applications [4].

ML is a technique that can forecast yields more accurately depending on a variety of factors. Learning is the main emphasis of this area of “Artificial Intelligence” (AI). By locating patterns as well as connections, ML can extract information from datasets. An ML model may be either descriptive or predictive, but it must be trained using datasets that show the results based on prior knowledge of the study problem and research goals.

The authors of this research used a decision tree “machine learning” method to predict crop production, which may be useful to farmers. Additionally, it is useful to make crop growth recommendations based on soil moisture information. Based on the characteristics of the soil, the model makes predictions and provides the most accurate crop. Before starting farming, this forecast will help the

farmers to estimate the production of the crop, enabling them to choose the best option. This method of machine learning provides a useful foundation for data-driven decision making. Using a collection of machine learning approaches, the program helps farmers produce crops based on key field characteristics [5]. In this research work, few machine learning algorithms are discussed in literature survey chapter, further we highlighted our proposed system in methodology chapter, and compared our analysis in results chapter.

II. LITERATURE REVIEW

As there are many ML algorithms based on various applications. In this chapter, we discuss machine learning algorithms used for agriculture applications. Artificial Neural Network (ANN) machine learning algorithm mainly works based on weight processing parameter. It mainly used as topological-based algorithm in a multi-layered and back-propagation weighted technique for the implementation of crop yield prediction based on quality of soil [6],[7]. But as this ANN algorithm is applicable to number of input attributes are very less, It does not have an automatic algorithm to predict the appropriate crop for sample facts space.

Researchers extended working on agriculture application using Support Vector Machine (SVM) algorithm. This algorithm is also known as support vector regression mainly used for crop prediction. The basic purpose of this approach is to use a radial-basis kernel function or a polynomial regression function to achieve a non-linear function. However, for the optimization of complicated problems, it wants to minimize the complexities of linear equations by embracing a broad input data set [8], [9].

“K-Nearest Neighbors” (K-NN) Machine learning algorithm is also used in agriculture application by predicting and analyzing crop yield sample data. It holds the previous sample space data for prediction of target value. By applying the distance measures between the train sample inputs and corresponding target data, the target results of k-neighbors values are found. But it does not perform training and optimization procedure for expectation of crop yield based on soil moisture values, so time and space complexity are high using this K-NN algorithm[10].

Based on the survey, observations & few limitations, we extended our research work for the prediction of crop yield based on soil moisture using decision tree machine learning algorithm for achieving better accuracy results.

A. Research Question

The goal of this Systematic Literature Review is to learn about the works which is published in the field of machine learning and crop production prediction. Studies have been studied from a variety of perspectives in order to gain understanding. These are as follows:

- In the literature, what ML algorithms have been used to estimate agricultural yields?
- Which features have been utilized in the literature to estimate agricultural yields using ML?
- Which proposed recognition evaluation parameters and methodologies have been explored in the research??
- What are the challenges in applying machine learning to estimate agricultural yields?

III. METHODOLOGY

In our proposed system, we use efficient machine learning algorithm for crop prediction system based on soil moisture nutrient values. The Figure 1 demonstrate the block diagram of proposed system:

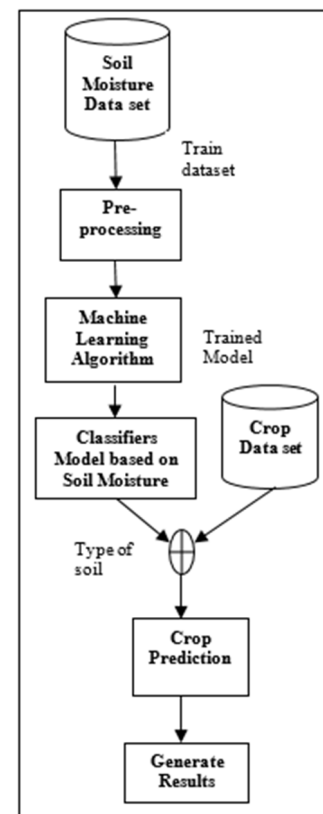


Figure 1: Block diagram of for crop prediction system based on soil moisture nutrient values.

In our proposed system, to determine the fertility of the given region, we know crops rely on different soil moisture data set. For example: Potatoes are most viable where the soil moisture pH level between 5.5 - 6.5 range, while Maize requires pH level between 6.0 - 7.5 range of optimum growth etc. Quality of the soil moisture can be examined using few important nutrient attributes present in the soil such as

Potassium (K), Phosphorus (P), Ca (Calcium), Nitrogen (N), pH value. The proposed system steps are as follows:

Step 1: Author used dataset as a sample: the first dataset is a soil moisture dataset (shown in below Table 1)

TABLE I. THE TABLE ILLUSTRATES THE SOIL MOISTURE DATA SET

N	P	K	PH
226.05	18	142	7.86847465
308.25	298	339.35	5.00530698
277.42	48	514.29	6.13613187
236.32	46	496.98	7.52408008
226.05	42	609.52	5.70694325
287.7	31	576.73	5.75850632
226.05	38	273.57	6.34960633
205.5	36	316.11	5.99996903
267.15	64	521.02	5.45359203
308.25	23	369.02	6.37257633
256.77	21	364.64	5.10558836
205.05	108	610.03	5.93534441
267.15	72	578.41	5.17782304
267.15	58	242.89	6.46239161
205.05	28	225.64	7.06917223
308.25	40	365.12	7.47313438
287.7	91	512.4	6.15837697
236.05	53	600.38	7.54987368
256.87	235	469.67	6.02832156
236.32	38	390.47	6.60499348

TABLE II. THE TABLE ILLUSTRATES THE CROP DATASET

Amount of "N"	Amount of "P"	Amount of "K"	Amount of "PH"	Crop
90	42	43	6.50298529	Rice
85	58	41	7.03809636	Rice
60	55	44	7.84020714	Rice
74	35	40	6.98040091	Rice
78	42	42	7.62847289	Rice
71	54	16	5.74991442	Maize
61	44	17	6.93175656	Maize
80	43	16	6.66796475	Maize
73	68	21	6.59606065	Maize
72	55	17	6.74991442	Maize
50	56	76	7.96660503	Chickpea
39	71	84	8.14082544	Chickpea
25	78	76	7.22896345	Chickpea
31	70	77	6.49254605	Chickpea
26	80	83	7.52859996	Chickpea

18	79	20	5.87734752	Kidney beans
21	63	17	5.97997397	Kidney beans
24	80	22	5.63599397	Kidney beans
34	60	22	5.63523178	Kidney beans
16	75	21	5.67922435	Kidney beans

The soil moisture dataset shown in Table 1 consists parameters like: Phosphorus (P), nitrogen (N), potassium (K), pH values dataset. And the crop dataset shown in Table 2 consists of important factors like: type of crop yield prediction like rice, maize, etc based on amount of N, P, K, pH values.

Step 2: The sample soil moisture and crop dataset is cleaned and noise data or missing data is been removed by converting into integer. In pre-processing step, the dataset is loaded to extract the noisy data, duplicate and replaces with the converted data into integers. This pre-processed data will be further stimulated to remove duplicate data from the dataset and will split the dataset into training and testing dataset. When splitting the dataset into test data and train pre-processed data, we consider test dataset as 0.2 (i.e., 20%) and train dataset as 0.9 (i.e., 90%).

Step 3: Datasets are in .CSV format and once we fetch or import the soil moisture dataset and crop dataset, further preprocessed, then our dataset is ready to train and test the model. We perform decision tree machine learning algorithm for our sample datasets to achieve effective prediction of crop yield. This decision tree algorithm forms many decision trees based on count value and maximum number of trees our algorithm measures accurately to provide same output results. We split the dataset in half, ninety percent for training and twenty percent for testing. The supervised learning algorithms family comprises the Decision Tree. The objective of using such a decision tree machine learning algorithms is to create a training approach to estimate the target variable's class or value based on fundamental decision learning rules from a previous dataset.. This algorithm observes the difference between expected outputs with the sample dataset and updates the weights between the nodes. This decision tree algorithm model is trained with much many iterations until we achieve desired output.

IV. RESULTS AND DISCUSSION

The research effort is carried out to predict the crop yield production by analyzing and processing the datasets by implementing in Python programming using Anaconda software. The results illustrated in figure 2:

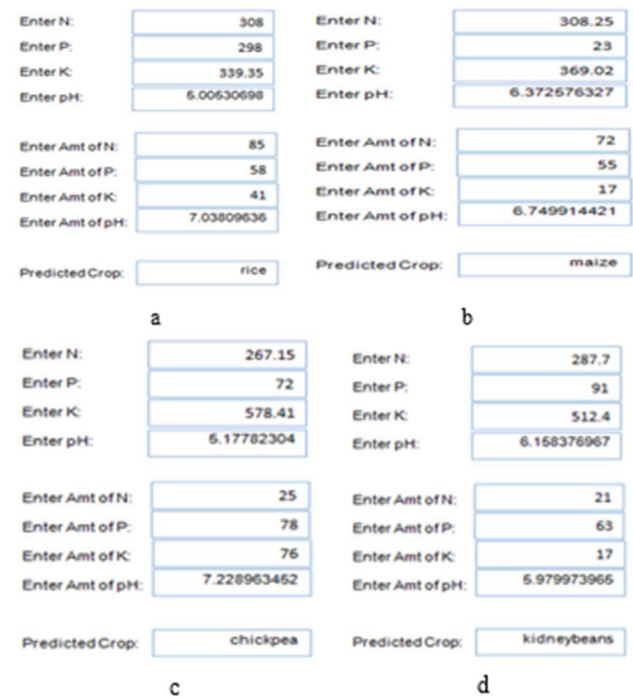


Figure 2: Results received using different inputs using python programming using anaconda software. (a) user input 1 - achieved predicted crop output, (b) user input 2 - achieved predicted crop output, (c) user Input 3 - achieved predicted crop output, (d) user input 4 - achieved predicted crop output.

Using machine learning decision tree algorithm for prediction of crop yield based on soil moisture data gives better results. We achieved performance measures error rate as 0.5 and accuracy as 95% in our proposed system is shown in below Figure 3:

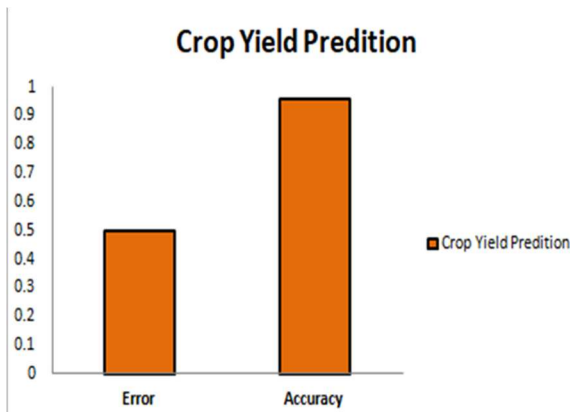


Figure 3: Performance measures: error and accuracy of crop yield prediction

- The system can be improved further by adding the following features:
- Image processing for crop disease detection, where users can upload a photo of a problematic crop and receive pesticide suggestions.

- Implementation of a Smart Irrigation System to automatically adjust watering schedules based on weather and soil conditions, plant water demand, and other factors.

V. CONCLUSION

The system employs both supervised and unsupervised Machine Learning algorithms to produce the best possible results in terms of accuracy. The method will aid farmers in overcoming their hardships and preventing suicide attempts. It will serve as a conduit for providing farmers with the information they need to achieve high yields and so optimize revenues, lowering suicide rates and making life easier for them. Farmers using this suggested strategy give priority to the crops that will produce the most on the land they own, resulting in maximum crop production. In India, 50 percent of the workforce is immediately employed in agriculture, while this sector only accounts for about 17 percent of the total employment. So, using this system, It will be directly or indirectly helping the economic level of farming in the country.

Based on the characteristics of the soil, the model makes predictions and provides the most accurate crop. Before starting farming, this forecast will help the farmers to estimate the production of the crop, enabling them to choose the best option. This method of machine learning provides a useful foundation for data-driven decision making. Using a collection of machine learning approaches, the program helps farmers produce crops based on key field characteristics. In this paper, Author proposed decision tree supervised machine learning algorithm to improve the results for the prediction of crop yield based on soil moisture parameters to achieve better error rate and accuracy for economic growth. It also includes the few machine learning algorithms which are discussed in literature survey, further Author highlighted the proposed system in methodology, and compared the analysis in results to give it a balance view. In future scope work for automatic detection of crop yield based on soil moisture data by considering more parameters like weather forecast, soil testing measures and so on can be used to help farmers for better growth of crop yield.

VI. ACKNOWLEDGMENTS

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VII. CONFLICT OF INTEREST

The authors declare that there is no conflict of interest among all the authors.

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